Master-slave Mechanism Synchronized by Three-phase Electrostatic Induction

Takuya Hosobata, Akio Yamamoto and Toshiro Higuchi
Department of Precision Engineering, School of Engineering
the University of Tokyo
7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

***Abstract*—In some environments, electromagnetic motors cannot be used because of the strong electromagnetic field they produce. Examples of such environments are inside magnetic resonance imaging (MRI) devices and inside vacuum chambers with presence of electron beams. In such cases, an electromagnetic motor can be replaced by a DEMED (Dual Excitation Multiphase Electrostatic Drive), which is a high power linear electrostatic motor, whose capability to operate in such environments has been proven by past studies.**

Since human hands cannot perform any task in these environments, they must be replaced by a master-slave system with manipulators controlled from outside the environment. However, to realize a master-slave system with an intuitive interface, the system must incorporate additional sensors and controllers, which makes the system more complicated.

A simple master-slave mechanism comprised of two DEMED type electrostatic motors is introduced in this paper. The two motors are passively synchronized by three-phase electrostatic induction, and thus no additional sensors are necessary. Feasibility of the synchronization was confirmed by an experiment, and the maximum attainable speed was 0.1 mm/s.

I. Introduction

Since the late 1980's, vast number of studies on electrostatic motors have been done in the field of MEMS, for this motor is believed to be most suitable in actuating mechanisms of microscopic size, utilizing the scaling effect of electrostatic force[1]. On the other hand, there are not many works on macroscale electrostatic motors, and one of the few successful examples is the DEMED (Dual Excitation Multiphase Electrostatic Drive) developed by Niino et al.[2], which realized a high power-to-weight ratio by integrating microscopic elements within flexible printed circuits. This work has indicated that electrostatic motors are also capable of being used in the macroscale applications.

Macroscale electrostatic motors are anticipated to be used in environments in which electromagnetic motors cannot be used, e.g. inside magnetic resonance imaging (MRI) devices[3], and inside vacuum chambers with presence of electron beams[4]. In these environments, direct operation by human hands either interferes with the process or is merely impossible, and therefore it must be replaced by a master-slave system, in which manipulators inside are controlled by an operator outside the environment. This master-slave system must be able to supply sufficient feedback to the operator, in order for its interface to be intuitive. To realize such a system, it must incorporate additional sensors and controllers, and therefore the system becomes complicated.

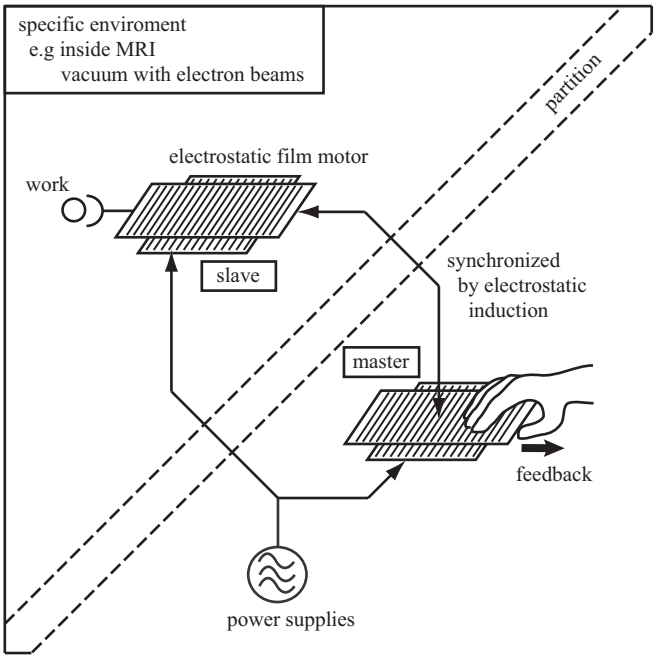


Fig. 1. Concept of the master-slave mechanism.

In this paper, we propose a master-slave mechanism comprised of two DEMED type electrostatic motors passively synchronized by three-phase electrostatic induction, developed with intention to be used in the applications mentioned above. The concept of this mechanism is shown in Fig. 1. A voltage induction type electrostatic motor was reported by Yamashita et al.[5], which has proved that the power for driving can be transmitted with no physical contact, in other words with no connection of the power supplies to the movable part of the motor. The master-slave mechanism also utilizes power transmission by electrostatic induction, by which synchronization of two motors with a simple configuration is realized. With this mechanism, applications such as remote manipulation of objects in vacuum enviroments and communication with patients inside MRI while sharing haptic information, are expected to be realized without complicated systems.

978-1-4244-4649-0/09/$25.00 ©2009 IEEE

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.

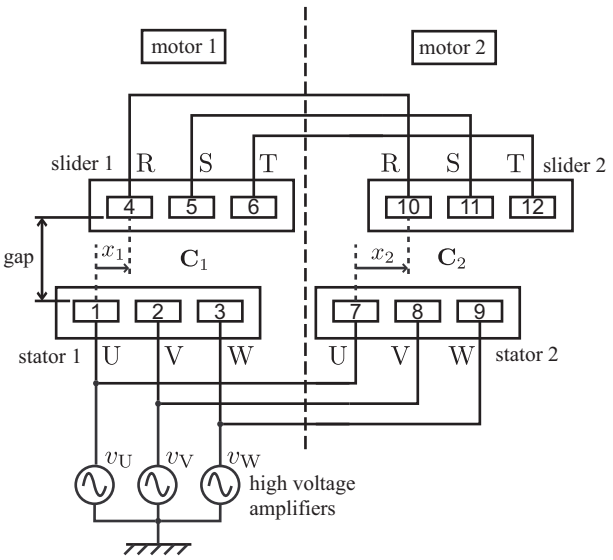


Fig. 2. Schematic diagram of the master-slave mechanism.

II. Theoretical Modeling

In this section, the modeling of the master-slave mechanism is discussed. An approximated model of the electrostatic motor was proposed by Yamamoto [6] based on a six-terminal capacitance network model. In the same paper he has also pointed out that when operated in single frequency mode, the six-terminal model can be converted into a three-terminal model, which can also be applied to the master-slave mechanism proposed in this present paper.

A. Three-terminal Capacitance Network Model

Fig. 2 shows a schematic diagram of the master-slave mechanism. It consists of two electrostatic motors, which will be referred to as motor 1 and motor 2 respectively in the further discussion. Each motor is comprised of two sets of three-phase electrodes, and the two sets are constrained to each other in the normal direction of the electrode surface with a constant gap, and have a single degree of freedom along the tangential direction (x_1 and x_2 direction for each corresponding motor). The fixed and the movable sets of three-phase electrodes will be referred to as stators and sliders respectively. Electrodes are numbered as shown in the figure.

The electrodes in the two stators are electrically connected to their counterparts, to which voltages are applied by high voltage amplifiers. The electrodes of the two sliders are also connected to their counterparts, in the same relative order as in the stator electrodes, but not connected to the power supplies. As a matter of convenience, groups of electrodes connected to electrodes 1 through 6 will be called U, V, W, R, S and T phase electrodes, respectively.

For each motor, we assume that six electrodes form a capacitance network represented by a 6-by-6 capacitance matrix $\mathbf{C}_i (i = 1, 2)$. Using these matrices, relations between voltages and induced charges of electrodes are given by the following:

$$\begin{aligned}
\mathbf{q}_1 &= \mathbf{C}_1 \mathbf{v}_1 \\
\mathbf{q}_2 &= \mathbf{C}_2 \mathbf{v}_2 \\
\mathbf{q}_1 &= \begin{bmatrix} q_1 & q_2 & q_3 & q_4 & q_5 & q_6 \end{bmatrix}^\mathrm{T} \\
\mathbf{q}_2 &= \begin{bmatrix} q_7 & q_8 & q_9 & q_{10} & q_{11} & q_{12} \end{bmatrix}^\mathrm{T} \\
\mathbf{v}_1 &= \begin{bmatrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{bmatrix}^\mathrm{T} \\
\mathbf{v}_2 &= \begin{bmatrix} v_7 & v_8 & v_9 & v_{10} & v_{11} & v_{12} \end{bmatrix}^\mathrm{T},
\end{aligned} \tag{1}$$

where q_j is the amount of induced charges and v_j is the voltage, respectively on electrode j ($j = 1, 2, \cdots 12$). Each capacitance matrix can be divided into four 3-by-3 submatrices:

$$\begin{aligned}
\mathbf{C}_i &= \begin{bmatrix} \mathbf{C}_{\mathrm{ST}i} & \mathbf{C}_{\mathrm{M}i} \\ \mathbf{C}_{\mathrm{M}i}^\mathrm{T} & \mathbf{C}_{\mathrm{SL}i} \end{bmatrix} \\
i &= 1, 2.
\end{aligned} \tag{2}$$

The matrices $\mathbf{C}_{\mathrm{ST}i}$ and $\mathbf{C}_{\mathrm{SL}i}$ are the sets of capacitance coefficients between two electrodes in the same group (stator and slider electrodes respecitively), whereas the matrix $\mathbf{C}_{\mathrm{M}i}$ is the set of coefficients between two electrodes from different groups: one from a stator and the other from a slider.

Voltages v_U, v_V and v_W are applied to U, V, W phase electrodes as shown in Fig.2, therefore $\mathbf{v}_1$ and $\mathbf{v}_2$ can be divided into two 3-by-1 voltage vectors:

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{v}_2 = \begin{bmatrix} \mathbf{v}_\mathrm{ST} \\ \mathbf{v}_\mathrm{SL} \end{bmatrix} \\
\mathbf{v}_\mathrm{ST} &= \begin{bmatrix} v_\mathrm{U} & v_\mathrm{V} & v_\mathrm{W} \end{bmatrix}^\mathrm{T} \\
\mathbf{v}_\mathrm{SL} &= \begin{bmatrix} v_\mathrm{R} & v_\mathrm{S} & v_\mathrm{T} \end{bmatrix}^\mathrm{T}
\end{aligned} \tag{3}$$

where v_R, v_S and v_T are the voltages induced on R, S and T phase electrodes respectively.

As for the induced charges, since R, S and T phase electrodes are not connected to the power supplies, total amount of charges induced must be equal to zero on each phase. Thus, charges induced on each phase are given as

$$\begin{aligned}
\mathbf{q}_1 + \mathbf{q}_2 &= \begin{bmatrix} \mathbf{q}_\mathrm{ST} \\ \mathbf{0} \end{bmatrix} \\
\mathbf{q}_\mathrm{ST} &= \begin{bmatrix} q_\mathrm{U} & q_\mathrm{V} & q_\mathrm{W} \end{bmatrix}^\mathrm{T},
\end{aligned} \tag{4}$$

where q_U, q_V and q_W are the charges induced on U, V and W phase electrodes respectively.

Substituting (1) and (3) into (4) gives

$$\mathbf{q}_\mathrm{ST} = (\mathbf{C}_{\mathrm{ST}1} + \mathbf{C}_{\mathrm{ST}2})\mathbf{v}_\mathrm{ST} + (\mathbf{C}_{\mathrm{M}1} + \mathbf{C}_{\mathrm{M}2})\mathbf{v}_\mathrm{SL}, \tag{5}$$

$$\mathbf{0} = (\mathbf{C}_{\mathrm{M}1}^\mathrm{T} + \mathbf{C}_{\mathrm{M}2}^\mathrm{T})\mathbf{v}_\mathrm{ST} + (\mathbf{C}_{\mathrm{SL}1} + \mathbf{C}_{\mathrm{SL}2})\mathbf{v}_\mathrm{SL}, \tag{6}$$

and by solving (6), induced voltages are obtained as

$$\mathbf{v}_\mathrm{SL} = -(\mathbf{C}_{\mathrm{SL}1} + \mathbf{C}_{\mathrm{SL}2})^{-1}(\mathbf{C}_{\mathrm{M}1}^\mathrm{T} + \mathbf{C}_{\mathrm{M}2}^\mathrm{T})\mathbf{v}_\mathrm{ST}. \tag{7}$$

Finally, by substituting (7) into (5), the three-terminal capacitance matrix model is derived as

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.

$$\mathbf{q}_{\mathrm{ST}} = \mathbf{C}^* \mathbf{v}_{\mathrm{ST}}, \tag{8}$$

where $\mathbf{C}^*$ is the 3-by-3 three-terminal capacitance matrix given as

$$\begin{aligned} \mathbf{C}^* &= \mathbf{C}^*_{\mathrm{ST}} - \mathbf{C}^*_{\mathrm{M}} \mathbf{C}^{*-1}_{\mathrm{SL}} \mathbf{C}^{*\mathrm{T}}_{\mathrm{M}} \\ \mathbf{C}^*_{\mathrm{ST}} &= \mathbf{C}_{\mathrm{ST1}} + \mathbf{C}_{\mathrm{ST2}} \\ \mathbf{C}^*_{\mathrm{SL}} &= \mathbf{C}_{\mathrm{SL1}} + \mathbf{C}_{\mathrm{SL2}} \\ \mathbf{C}^*_{\mathrm{M}} &= \mathbf{C}_{\mathrm{M1}} + \mathbf{C}_{\mathrm{M2}}. \end{aligned} \tag{9}$$

From this model, thrust force of each motor f_i can simply be obtained by taking partial derivatives of the system's energy, in accordance to the virtual work theory.

$$\begin{aligned} f_i &= \frac{1}{2} \frac{\partial}{\partial x_i} (\mathbf{v}^{\mathrm{T}}_{\mathrm{ST}} \mathbf{C}^* \mathbf{v}_{\mathrm{ST}}) \\ i &= 1, 2 \end{aligned} \tag{10}$$

B. Simplified Model

The elements of the submatrices in (2) are dependent on the relative displacement between a slider and a stator, which can be approximated as a periodic function of x_i with a fundamental period of 3 electrode pitches and its harmonics [6]. Simplified submatrices neglecting 2nd and higher order harmonics are given as

$$\begin{aligned} \mathbf{C}_{\mathrm{ST}i} &= \begin{bmatrix} C_{\mathrm{ST}i} & -C_{\mathrm{T}i} & -C_{\mathrm{T}i} \\ -C_{\mathrm{T}i} & C_{\mathrm{ST}i} & -C_{\mathrm{T}i} \\ -C_{\mathrm{T}i} & -C_{\mathrm{T}i} & C_{\mathrm{ST}i} \end{bmatrix} \\ \mathbf{C}_{\mathrm{SL}i} &= \begin{bmatrix} C_{\mathrm{SL}i} & -C_{\mathrm{L}i} & -C_{\mathrm{L}i} \\ -C_{\mathrm{L}i} & C_{\mathrm{SL}i} & -C_{\mathrm{L}i} \\ -C_{\mathrm{L}i} & -C_{\mathrm{L}i} & C_{\mathrm{SL}i} \end{bmatrix} \\ \mathbf{C}_{\mathrm{M}i} &= \begin{bmatrix} C^0_{\mathrm{M}i} & C^+_{\mathrm{M}i} & C^-_{\mathrm{M}i} \\ C^-_{\mathrm{M}i} & C^0_{\mathrm{M}i} & C^+_{\mathrm{M}i} \\ C^+_{\mathrm{M}i} & C^-_{\mathrm{M}i} & C^0_{\mathrm{M}i} \end{bmatrix} \\ C^0_{\mathrm{M}i} &= -C_{\mathrm{M0}i} - C_{\mathrm{MA}i} \cos(\frac{2\pi}{3p_i} x_i) \\ C^+_{\mathrm{M}i} &= -C_{\mathrm{M0}i} - C_{\mathrm{MA}i} \cos(\frac{2\pi}{3p_i} x_i + \frac{2}{3}\pi) \\ C^-_{\mathrm{M}i} &= -C_{\mathrm{M0}i} - C_{\mathrm{MA}i} \cos(\frac{2\pi}{3p_i} x_i - \frac{2}{3}\pi) \\ i &= 1, 2. \end{aligned} \tag{11}$$

The coefficients $C_{\mathrm{ST}i}$ and $C_{\mathrm{SL}i}$ are self-induction coefficients of electrodes, $C_{\mathrm{T}i}$ and $C_{\mathrm{L}i}$ are induction coefficients between pairs of neighboring electrodes, and $C_{\mathrm{M0}i}$ and $C_{\mathrm{MA}i}$ are fourier components of the induction coefficients between opposing pairs of electrodes. All of these coeffients are positive and constant parameters to be identified by experiments. The constant p_i is the electrode pitch of each corresponding motor.

Here we define the applied voltages as:

$$\begin{aligned} v_{\mathrm{U}} &= 0 \\ v_{\mathrm{V}} &= -v_0 \\ v_{\mathrm{W}} &= v_0, \end{aligned} \tag{12}$$

where v_0 is the voltage amplitude.

By substituting (11) and (12) into (7), we have a simplified form of induced voltages on the slider electrodes:

$$\begin{aligned} v_{\mathrm{R}} &= -\frac{\sqrt{3} v_0}{C'} (C_{\mathrm{MA1}} \sin\theta_1 + C_{\mathrm{MA2}} \sin\theta_2) \\ v_{\mathrm{S}} &= -\frac{\sqrt{3} v_0}{C'} \{C_{\mathrm{MA1}} \sin(\theta_1 + \frac{2}{3}\pi) + C_{\mathrm{MA2}} \sin(\theta_2 + \frac{2}{3}\pi)\} \\ v_{\mathrm{T}} &= -\frac{\sqrt{3} v_0}{C'} \{C_{\mathrm{MA1}} \sin(\theta_1 - \frac{2}{3}\pi) + C_{\mathrm{MA2}} \sin(\theta_2 - \frac{2}{3}\pi)\}, \end{aligned} \tag{13}$$

where C' is the sum of coefficients and θ_i is the positional phase of the slider i, which are defined as

$$\begin{aligned} C' &= C_{\mathrm{SL1}} + C_{\mathrm{L1}} + C_{\mathrm{SL2}} + C_{\mathrm{L2}} \\ \theta_i &= \frac{2\pi}{3p_i} x_i \\ i &= 1, 2. \end{aligned} \tag{14}$$

By substituting (11) and (12) into (10), simplified expressions of the thrust forces are derived as

$$\begin{aligned} f_1 &= -\frac{3\pi}{p_1} \frac{C_{\mathrm{MA1}} C_{\mathrm{MA2}}}{C'} v_0^2 \sin(\theta_1 - \theta_2 \pm \pi) \\ f_2 &= -\frac{3\pi}{p_2} \frac{C_{\mathrm{MA1}} C_{\mathrm{MA2}}}{C'} v_0^2 \sin(\theta_2 - \theta_1 \mp \pi). \end{aligned} \tag{15}$$

From (15), we can see that the thrust force of each motor is determined by the difference of the positional phases of the two sliders, with a phase offset of $\pm\pi$. The phase offset was added to make the coefficient negative, indicating that the force is a restoring force around the equilibrium point. In equilibrium state, when no loads are applied to the sliders, the relation between positions of the two sliders is obviously given by

$$x_1 = \frac{p_1}{p_2} x_2 \mp \frac{3}{2} p_1, \tag{16}$$

which means that motions of the two sliders are synchronized: motion of one slider will be reproduced in the other with a ratio determined by their electrode pitch. As for a case when a certain load is applied to one slider, here we suppose it is slider 2, thrust force of slider 1 will be

$$f_1 = -\frac{p_2}{p_1} f_2, \tag{17}$$

meaning that applied force to one slider is reproduced on the other, reduced or amplified by the reciprocal of the ratio in (16).

From this model, it is expected that a simple configuration of master-slave mechanism without sensors for force or position feedback can be realized.

III. Experimental Results

In this section, the results of the experiments that were carried out using the setup shown in Fig. 3 are described.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.

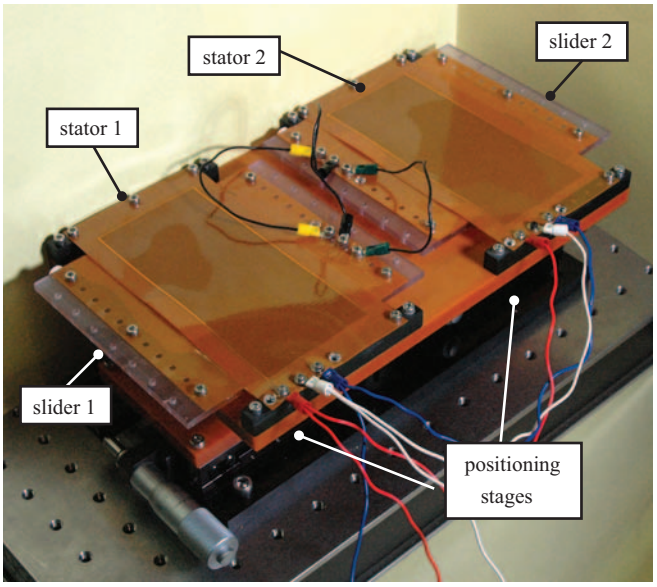


Fig. 3. Photograph of the experimental setup.

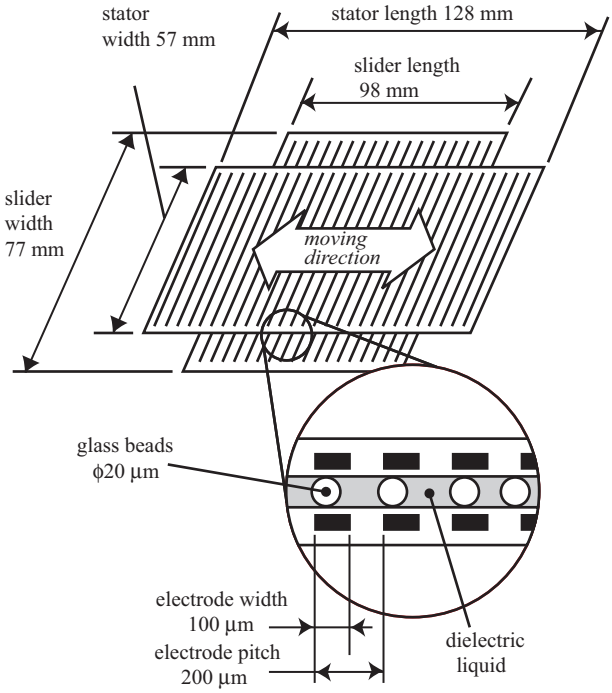


Fig. 4. Dimensions of an electrostatic film motor.

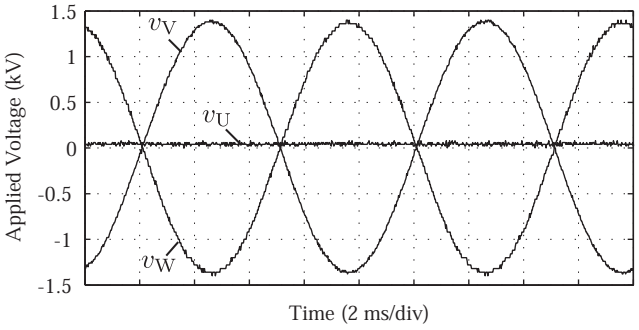


Fig. 5. Voltages applied to the stator electrodes.

A. Parameter Identification

Two identical motors, pairs of stator and slider electrodes fabricated on flexible printed circuits, were installed in the setup, of which dimensions are shown in Fig. 4. The electrode pitch was 200 μm and the effective electrode area (in other words overlapping area of a motor) was 5.6×10^{-3} m^2. Between the stator and slider electrodes, glass beads of 20 μm diameter were scattered to maintain a constant gap, and dielectric liquid *Fluorinert* FC-77(3M) was injected to prevent corona discharge resulting from the high voltages.

The capacitance coefficients of this motor were measured by the method proposed by Yamamoto[6] using an LCZ meter 2330 (NF Electronic Instruments), and the results were as follows.

$$\begin{aligned} C_{\mathrm{ST}i} &= 1390\ \mathrm{pF}, \quad C_{\mathrm{T}i} = 487\ \mathrm{pF} \\ C_{\mathrm{SL}i} &= 1350\ \mathrm{pF}, \quad C_{\mathrm{L}i} = 443\ \mathrm{pF} \\ C_{\mathrm{M0}i} &= 126\ \mathrm{pF}, \quad C_{\mathrm{MA}i} = 81\ \mathrm{pF} \\ i &= 1, 2 \end{aligned} \tag{18}$$

B. Applied and Induced Voltages

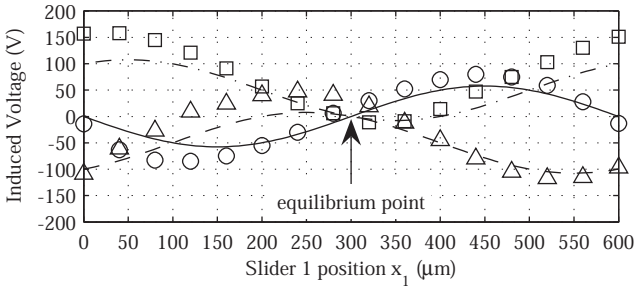
From the model described in the last section, one might conclude that applying DC voltages is sufficient to realize the synchronization of the two sliders. However, as was pointed out by Niino et al., thrust force characteristics of DEMED type electrostatic film motors deteriorate when it is operated under low frequency voltages[7], and the master-slave mechanism was found to be no exception, unable to operate under DC voltages. To solve the low frequency operation problem, amplitude-modulation method was proposed[7], which was also adopted in this experiment. Amplitude-modulated voltages are given as

$$\begin{aligned} v_{\mathrm{U}} &= 0 \\ v_{\mathrm{V}} &= -v_0 \cos 2\pi f_{\mathrm{m}} t \\ v_{\mathrm{W}} &= v_0 \cos 2\pi f_{\mathrm{m}} t, \end{aligned} \tag{19}$$

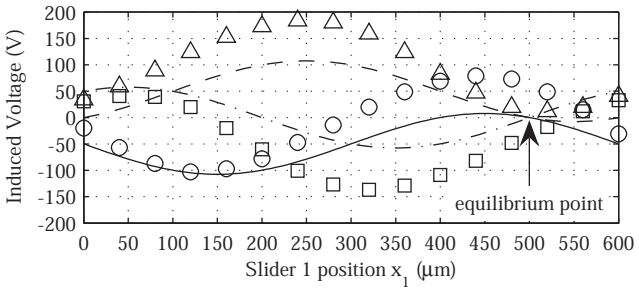
where f_{m} is the modulation frequency. Fig. 5 shows the voltages applied to U, V, and W phase electrodes in this experiment. Amplitude v_0 was set to 1.47 kV, modulated with frequency of 100 Hz.

Fig. 6 shows the amplitude of voltages induced on R, S and T phase electrodes, with different slider positions. Data shown in lines are calculated values by (13) using identified parameters in (18), and data shown in points are measured

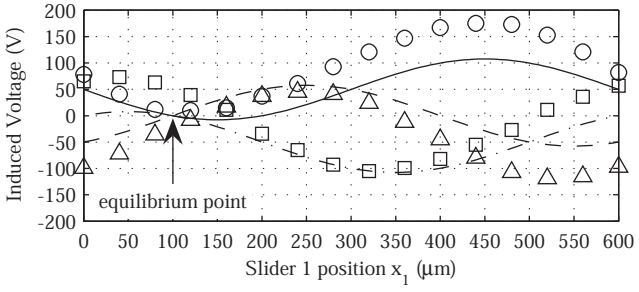
Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.



(a) Slider 2 Position $x_2 = 0$ μm



(b) Slider 2 Position $x_2 = 200$ μm



(c) Slider 2 Position $x_2 = 400$ μm

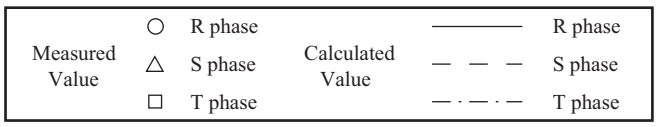


Fig. 6. Voltages induced on the stator electrodes.

values by oscilloscope. Equilibrium points are also shown in the figures.

From these graphs, we can see that although absolute values of the amplitudes are showing some errors, trend of the data are in good agreement with the theoretical model. The differences in the induced voltages of R, S and T phases increase as the slider 1 moves away from the equilibrium position, which results in producing a restoring force as shown in (15). The equilibrium position of a slider changes as the other slider moves, and thus the synchronization is made possible.

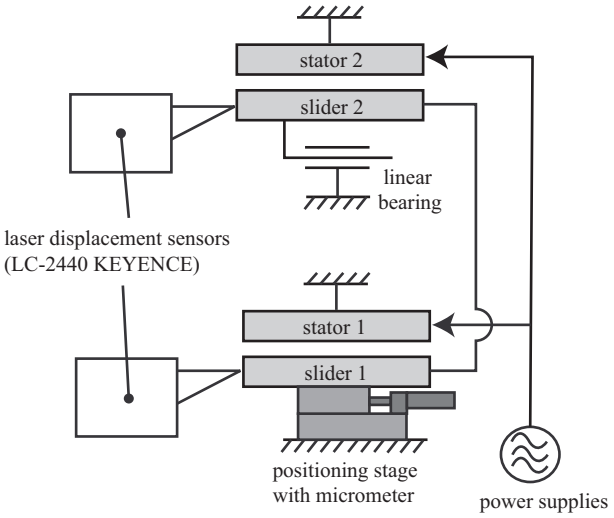


Fig. 7. Schematic diagram of the synchronization experiment.

C. Synchronization of the Two Sliders

Finally, an experiment was conducted using the setup shown in Fig. 7, to confirm the feasibility of the synchronization of the two sliders. The slider 2 was mounted on a linear ball bearing, free to move along the x_2 axis. On the other hand, the slider 1 was mounted on a micrometer positioning stage, and its position could manually be adjusted. Positions of the two sliders were measured by two laser displacement sensors, and the measured data were recorded on a PC via a DSP board.

The result is shown in Fig. 8. The solid line represents the trajectory of the slider 1, in this case the master slider, and the dotted line shows the trajectory of the slider 2, the slave slider. The slider 1 was moved back and forth within ± 2 mm stroke, slowly with a speed of about 0.1 mm/s. From this graph we can conclude that the slave slider is successfully following the trajectory of the master slider.

In this experiment, 0.1 mm/s was the maximum attainable speed, and the slave slider could not follow the master slider when the input speed exceeded this limit. One possible cause of this limitation is the friction between the stator and the slider of the slave motor: thrust force may not have been large enough to overcome the frictional force above this limit.

The same result from Fig. 8 is plotted in Fig. 9, in position vs. position form. From this graph, difference between forward and backward paths is observed. Apparently, the slider 2 does not start moving just after the change in moving direction of the slider 1. The slider 2 stays in its position until the slider 1 has moved about 150 μm, which also indicates the presence of the friction. The equation (15) indicates that the thrust force is zero at the equilibrium point, and thus certain displacement is needed to obtain a force large enough to overcome the static friction force. To clarify the cause of this hysteresis and also the cause of the previously mentioned speed limit, further study on thrust force characteristic is necessary.

Also from Fig. 9, spatially periodic wavering in both the

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.

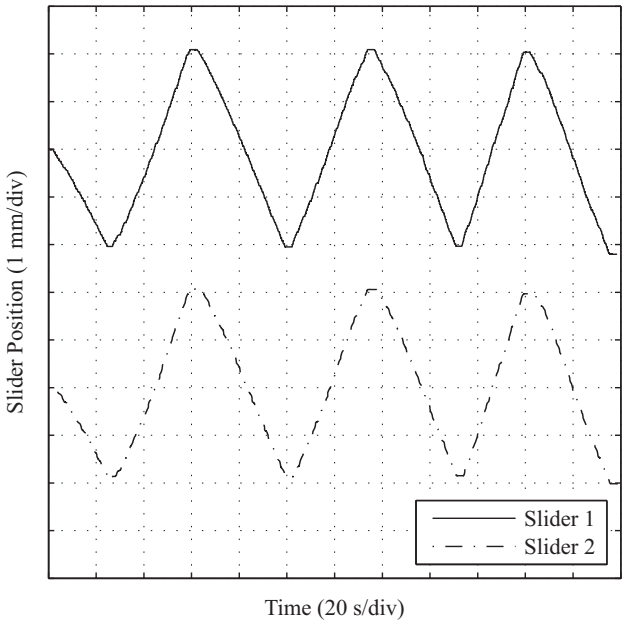


Fig. 8. Motion trajectory of the two sliders in time vs. position form.

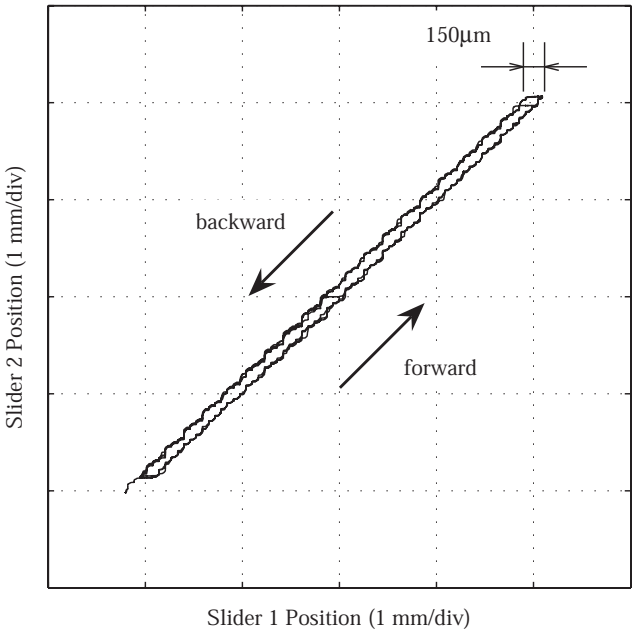


Fig. 9. Motion trajectory of the two sliders in position vs. position form.

forward path and the backward path is observed, apparently with a period of 200 μm, which is equal to the electrode pitch. This indicates that the thrust force will also have a non-linear characteristic, which will be a major problem when used in applications that require accurate force reproduction. Possible solution to this problem is the use of skewed electrodes, which is an already established method to reduce force ripples in the DEMED type electrostatic motors[8].

IV. Conclusion

A simple master-slave mechanism comprised of two DEMED type electrostatic motors, passively synchronized by three-phase electrostatic induction was presented. The master-slave mechanism was modeled as a three-terminal capacitance network, and the principle of the synchronization was explained based on this model.

Feasibility of the synchronization was confirmed by an experiment, showing good agreement with the theory. In the experiment, maximum attainable speed was 0.1 mm/s, which is not likely to be sufficient for practical use. Moreover, a hysteresis characteristic was observed, indicating that thrust force was not large enough to overcome the frictional force. In order for this mechanism to be utilized in practical applications, further study in the thrust force characteristic is necessary, along with the considerations on the thrust force enhancement.

With further improvements, this mechanism would be utilized as a component of a master-slave system with a haptic interface, in enviroments where electromagnetic motors cannot be used: e.g. inside MRI devices and inside vacuum chambers with presence of electron beams.

Acknowledgment

One of the authors (T. Hosobata) was supported through the Global COE Program, “Global Center of Excellence for Mechanical Systems Innovation,” by the Ministry of Education, Culture, Sports, Science and Technology.

References

[1] Long-Shen Fan, Yu-Chong Tai and Richard S. Muller, ”IC-processed electrostatic micromotors”, *Sensors and Actuators*, Vol. 20, No. 1-2, 1989, pp.41-47.

[2] T. Niino, T. Higuchi, and S. Egawa, ”Dual excitation multiphase electrostatic drive,” *Industry Applications Conference, 1995. Thirtieth IAS Annual Meeting, IAS '95., Conference Record of the 1995 IEEE*, Vol.2, pp.1318-1325 vol.2, 8-12 Oct 1995

[3] M. Rajendra, A. Yamamoto, T. Oda, H. Kataoka, H. Yokota, R. Himeno, and T. Higuchi, ”Motion Generation in MR Environment Using Electrostatic Film Motor for Motion-Triggered Cine-MRI,” *IEEE/ASME Transactions on Mechatronics*, Vol. 13, No. 3, pp.278-285, June 2008

[4] A. Yamamoto, H. Yasui, N. Shimizu and T. Higuchi, ”Development of electrostatic levitation motor for vacuum condition”, *2003 IEEE International Symposium on Industrial Electronics*, Vol. 2, pp.934-9, 2003

[5] Norio Yamashita, Zu Guang Zhang, Akio Yamamoto, Masahiko Gondo and Toshiro Higuchi, ”Voltage-induction type electrostatic film motor driven by two- to four-phase ac voltage and electrostatic induction”, *Sensors and Actuators A: Physical*, Vol. 140, No. 2, 10 November 2007, pp.239-250

[6] Akio Yamamoto, Toshiki Niino and Toshiro Higuchi, ”Modeling and identification of an electrostatic motor”, *Precision Engineering*, Vol. 30, No. 1, January 2006

[7] Toshiki Niino, Akio Yamamoto and Toshiro Higuchi, ”Operation of a dual excitation multiphase electrostatic drive by amplitude-modulated ac voltage”, *Electrical Engineering in Japan*, Vol. 131, No. 4, pp.78-84, 2000

[8] Akio Yamamoto, Toshiki Niino, Takahisa Ban, Toshiro Higuchi, ”A high-power electrostatic motor using skewed electrodes”, *Electrical Engineering in Japan*, Vol. 125, No.3, pp.50-58, 1998

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:15:27 UTC from IEEE Xplore. Restrictions apply.